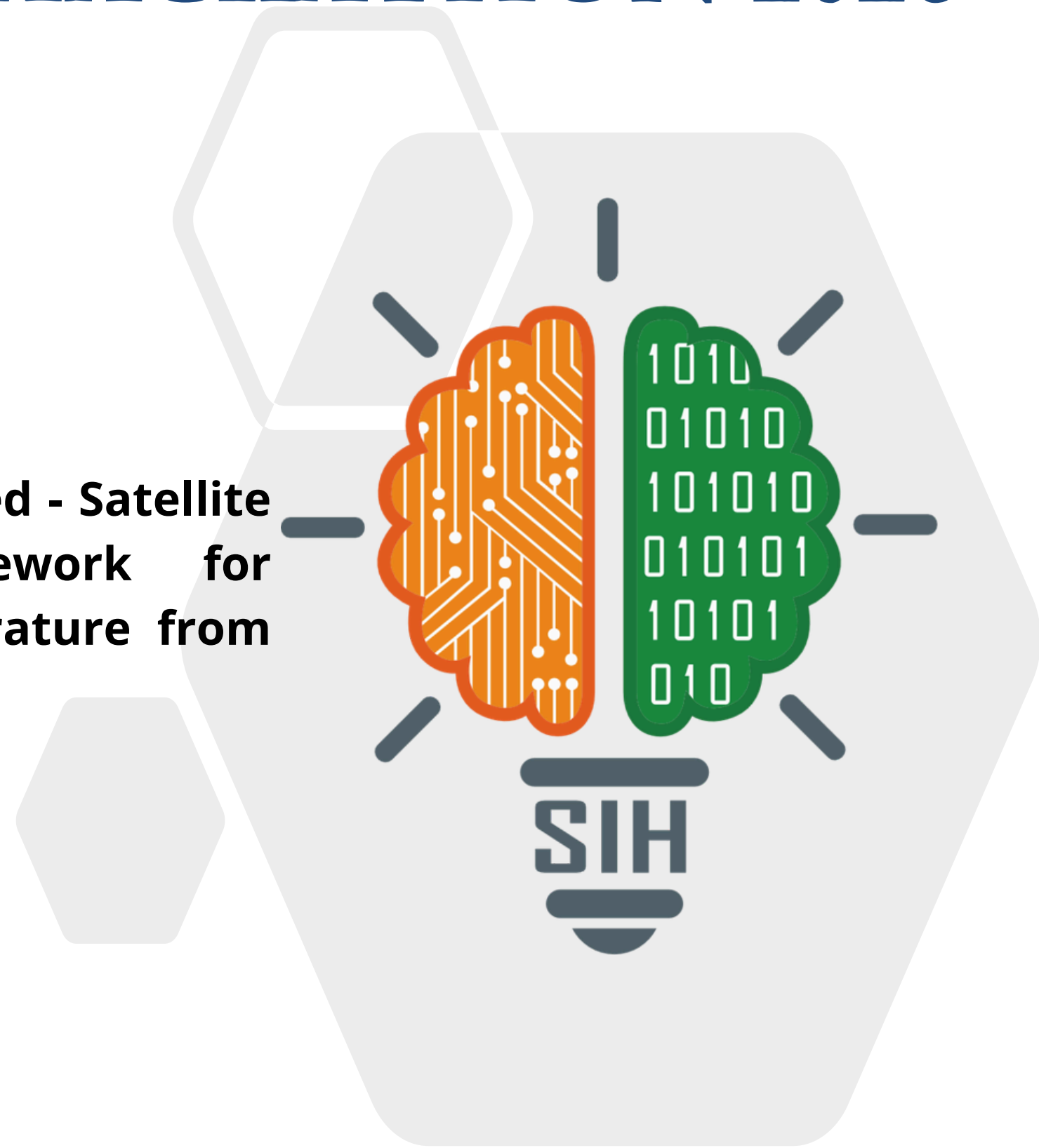


SMART INDIA HACKATHON 2026

- **Problem Statement ID – SIH26066**
- **Problem Statement Title-** OceanEmbed - Satellite Embedding-Based Deep Learning Framework for Reconstruction of Subsurface Ocean Temperature from Surface Satellite Observations.
- **Theme- Disaster Management**
- **PS Category- Software**
- **Team ID-**
- **Team Name- REGALIA**



OceanEmbed - Satellite Embedding-Based Deep Learning Framework for Reconstruction of Subsurface Ocean Temperature from Surface Satellite Observations.

What exists today?



ARGO – In-situ truth

Provides accurate temperature and salinity at different depths, but only at limited locations and times.



COPERNICUS – Surface view

Gives wide and frequent coverage, but mainly reflects what's happening at the ocean surface.



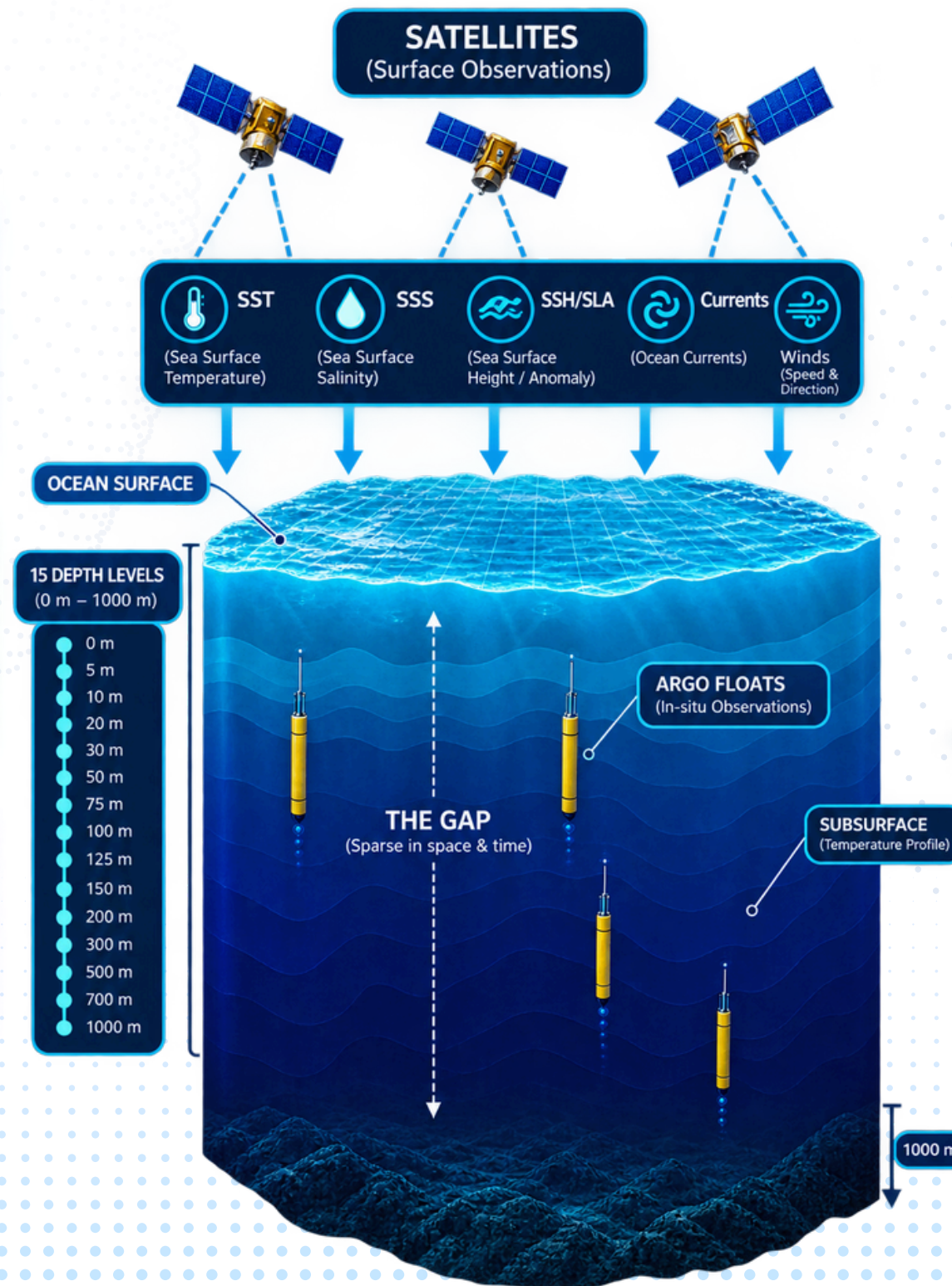
GLORYS – Reanalysis

Provides a complete 3D ocean picture, but is model-based reconstruction rather than direct real-time observation.



THE GAP

We have accurate measurements, wide surface coverage and model-based 3D data - but continuous real-time subsurface information is still missing.



What is our solution?

1

Surface-to-Subsurface Intelligence



- Use easily available satellite surface signals and learn their relationship with subsurface ocean conditions.
- Reconstruct temperature at depths where direct measurements are unavailable.

2

Multi-Source Ocean Data Fusion



- Combine Satellite data with Argo observations and reanalysis data for training and validation.
- Delivers a more reliable and continuous picture of the ocean beneath the surface.

DATA SOURCES

Multi-source ocean & atmospheric data

-  **CMEMS Altimetry**
SLA, ADT, UGOS, VGOS
-  **CMEMS SSS**
Sea Surface Salinity
-  **ERA5**
10 m U/V Winds
-  **NOAA OISST**
Sea Surface Temperature
-  **GLORYS12v1**
Subsurface Temperature (target)
-  **ARGO**
Independent validation profiles

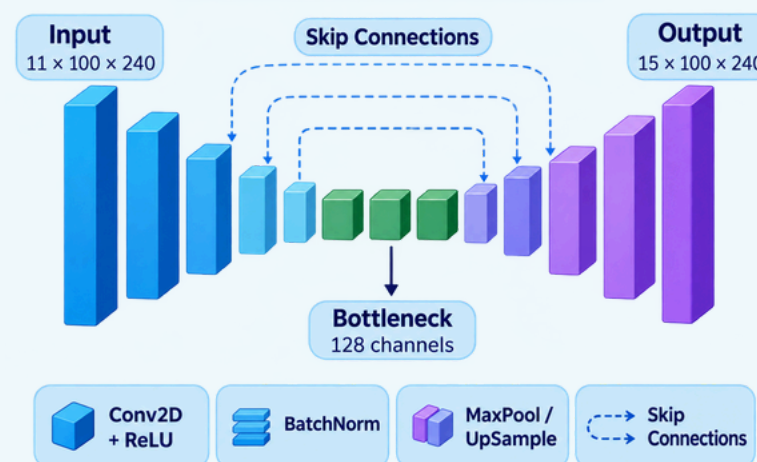
DATA PROCESSING

Spatially and temporally aligned gridded dataset

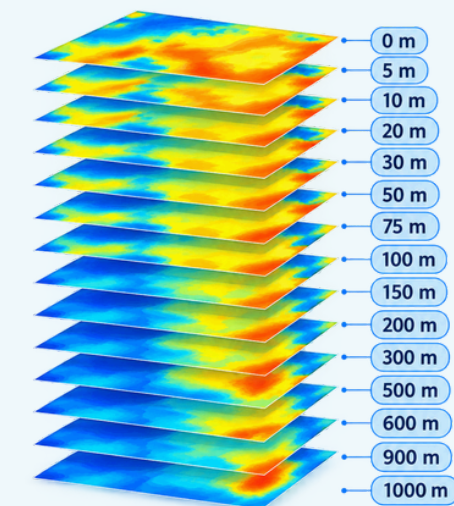
-  **Region of interest**
(5°–30°N, 45°–105°E)
-  **Common grid**
0.25° × 0.25°
(100 × 240)
-  **Temporal alignment**
(Daily aggregation)
-  **Interpolation & quality control**
-  **Normalization**
(per channel)

MODEL ARCHITECTURE

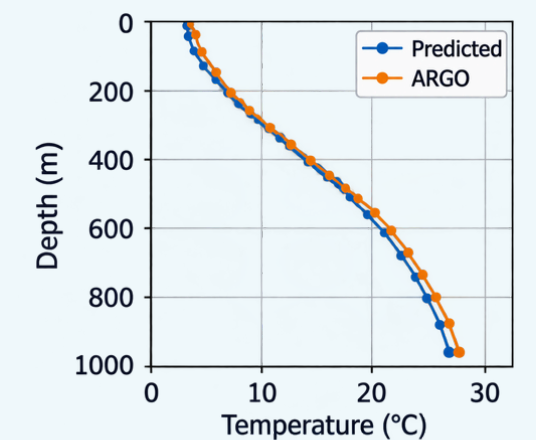
U-Net encoder-decoder with skip connections



PREDICTION OUTPUT

3D temperature field
(15 depth levels)

VALIDATION



- ✓ RMSE
- ✓ MAE
- ✓ R²
- ✓ Depth-wise error analysis
- ✓ Spatial validation



Programming

Python (NumPy, Xarray, Pandas, NetCDF4, SciPy)



Deep Learning

PyTorch (CNN, U-Net, DataLoader)



Training

Adam Optimizer, MSE Loss, Early Stopping, Batch Normalization, CUDA / GPU



Visualization

Matplotlib, Cartopy

Deployment
(Proposed)FastAPI (Backend), Web Frontend
Interactive Maps & Vertical Profiles

Unique Strengths



Multi-source, physically meaningful inputs

3D subsurface temperature prediction
from only surface observations

Independent validation using real-world ARGO profiles



Scalable and extensible for operational use

Supports ocean monitoring, climate research
and policy decision-making

01 Satellite Data Gaps**⚠️ RISK**

Clouds, sensor limitations, and retrieval errors may cause missing or noisy surface observations.

✅ MITIGATION

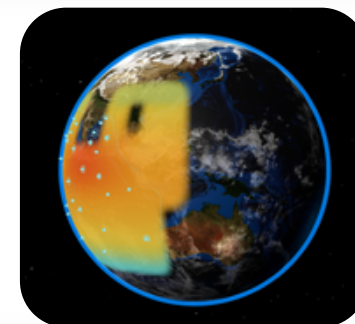
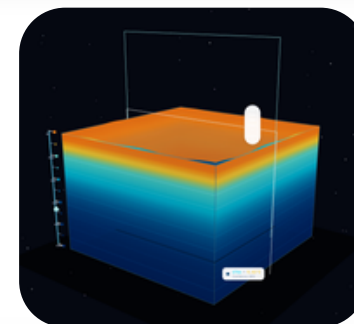
Fuse multiple sources with quality control, alignment, and missing-data handling before model input.

02 ARGO Observations are Sparse / Can Have Errors**⚠️ RISK**

ARGO measurements are spatially sparse and may contain uncertainty.

✅ MITIGATION

Keep ARGO independent for validation, using multiple quality-controlled profiles and statistical error metrics.

**03 Surface Data ≠ Subsurface Temperature****⚠️ RISK**

Surface temperature alone may not uniquely predict subsurface temperature.

✅ MITIGATION

Fuse SST, SSS, SLA/ADT, wind, chlorophyll, and spatial-temporal data to capture subsurface variability.

04 Different Datasets → Different Resolutions**⚠️ RISK**

Different sources have varying grids, time scales, units, and missing-data patterns

✅ MITIGATION

Standardize all sources to a $0.25^\circ \times 0.25^\circ$ daily grid using a unified preprocessing pipeline.

05 Reanalysis / Model Bias**⚠️ RISK**

GLORYS may contain model biases that could be transferred during training.

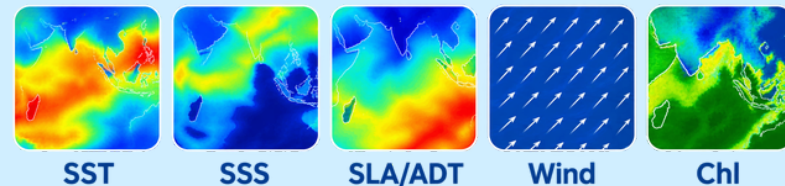
✅ MITIGATION

GLORYS provides the dense training reference, while independent ARGO observations validate predictions.

01 WHAT OUR MODEL DOES

**Multiple Surface Inputs**

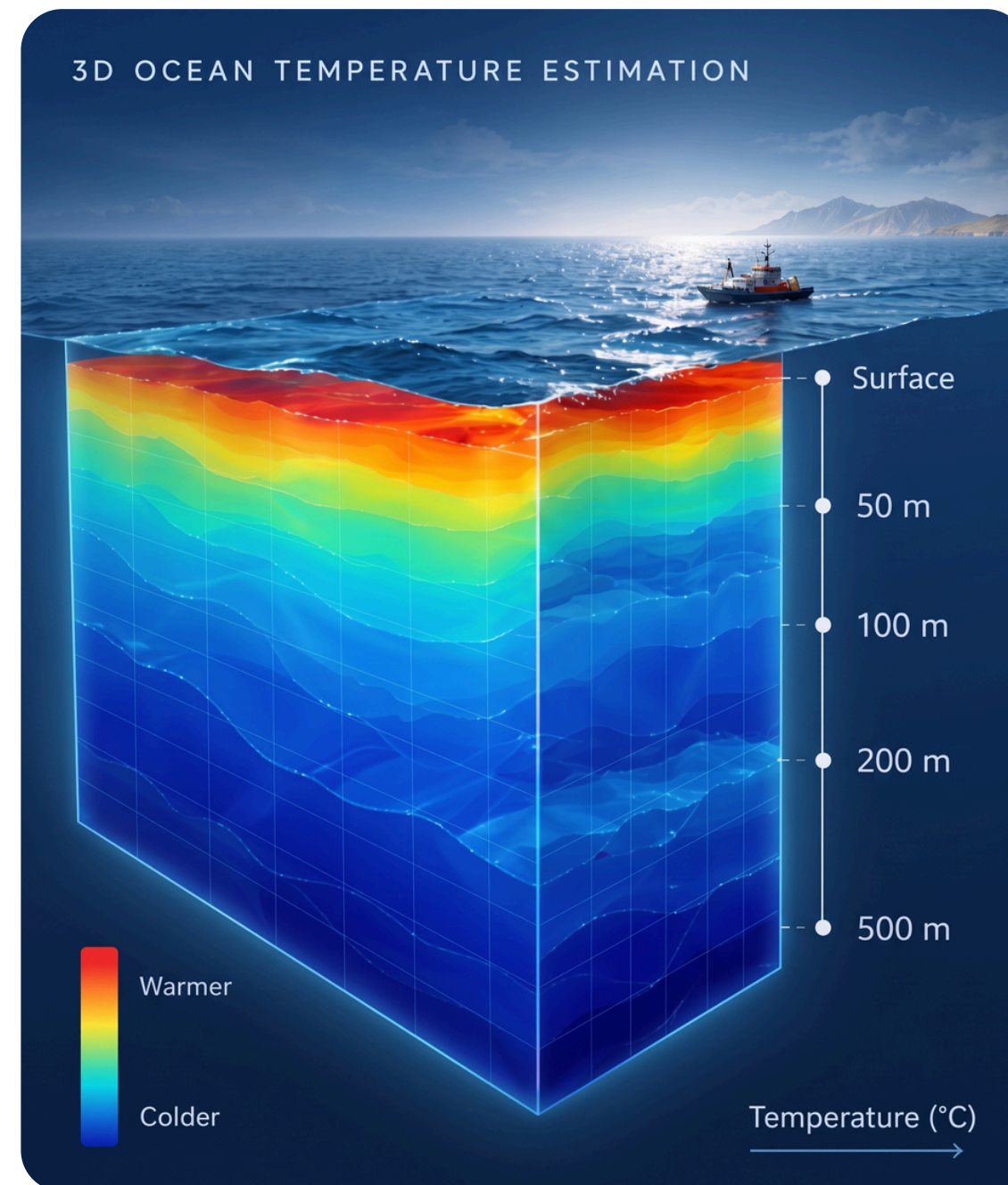
SST + SSS + SLA + WIND + CHLOROPHYLL

**Deep Learning Fusion**

Learns complex relationships between surface conditions and subsurface structure

**3D Temperature Reconstruction**

Daily temperature profiles at multiple depths, at 0.25° resolution



THE ACTUAL CONTRIBUTION OF OUR MODEL

02 KEY IMPACTS AND BENEFITS

**3D Ocean Visibility**

Estimates temperature beneath the surface.

1

**Cyclone Monitoring**

Reveals subsurface heat that can influence cyclone intensity.

2

**Marine Heatwave Detection**

Identifies warming hidden below the surface.

3

**Fisheries & Ecosystems**

Provides depth-wise thermal information for marine habitats.

4

**Better Ocean Forecasting**

Adds continuous 3D temperature information to forecasting systems.

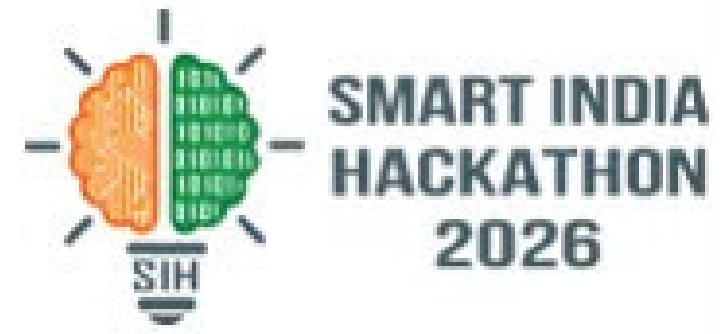
5

**High-Resolution Monitoring**

Daily temperature estimates at *0.25° resolution* across the North Indian Ocean.

6

RESEARCH AND REFERENCES



- research links